

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/Nassau Candy Distributor.csv')
```

```
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province	Postal Code	Division	Region	Product ID
0	1	US-2021-103800-CHO-MIL-31000	03-01-2024	30-06-2026	Standard Class	103800	United States	Houston	Texas	77095	Chocolate	Interior	CHO-MIL-31000
1	2	US-2021-112326-CHO-TRI-54000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-TRI-54000
2	3	US-2021-112326-CHO-NUT-13000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-NUT-13000
3	4	US-2021-112326-CHO-SCR-58000	04-01-2024	01-07-2026	Standard Class	112326	United States	Naperville	Illinois	60540	Chocolate	Interior	CHO-SCR-58000
4	5	US-2021-141817-CHO-TRI-54000	05-01-2024	05-07-2026	Standard Class	141817	United States	Philadelphia	Pennsylvania	19143	Chocolate	Atlantic	CHO-TRI-54000

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 18 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Row ID              10194 non-null  int64
1   Order ID            10194 non-null  object
2   Order Date          10194 non-null  object
3   Ship Date           10194 non-null  datetime64[ns]
4   Ship Mode           10194 non-null  object
5   Customer ID         10194 non-null  int64
6   Country/Region      10194 non-null  object
7   City                10194 non-null  object
8   State/Province      10194 non-null  object
9   Postal Code         10194 non-null  object
10  Division            10194 non-null  object
11  Region              10194 non-null  object
12  Product ID          10194 non-null  object
13  Product Name        10194 non-null  object
14  Sales               10194 non-null  float64
15  Units               10194 non-null  int64
16  Gross Profit        10194 non-null  float64
17  Cost                10194 non-null  float64
dtypes: datetime64[ns](1), float64(3), int64(3), object(11)
memory usage: 1.4+ MB
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'], format='%d-%m-%Y', errors='coerce')
df['Ship Date'] = pd.to_datetime(df['Ship Date'], format='%d-%m-%Y', errors='coerce')
```

```
df = df.dropna(subset=['Order Date', 'Ship Date'])
```

```
df['Lead Time'] = (df['Ship Date'] - df['Order Date']).dt.days
```

```
initial_count = len(df)
df = df[df['Lead Time'] >= 0]
dropped_negatives = initial_count - len(df)
```

```
df['Factory Location'] = 'Central Hub'
```

```
df['Route_Region'] = df['Factory Location'] + ' -> ' + df['Region']
df['Route_State'] = df['Factory Location'] + ' -> ' + df['State/Province']
```

```
route_agg = df.groupby('Route_State').agg(
    Total_Shipments=('Lead Time', 'count'),
    Avg_Lead_Time=('Lead Time', 'mean'),
    Lead_Time_Std=('Lead Time', 'std')
).reset_index()
route_agg['Lead_Time_Variability'] = route_agg['Lead_Time_Std'] / route_agg['Avg_Lead_Time']
```

```
route_agg_sorted = route_agg.sort_values('Avg_Lead_Time')
top_10 = route_agg_sorted.head(10)
bottom_10 = route_agg_sorted.tail(10)
ship_mode_perf = df.groupby('Ship Mode').agg(
    Total_Shipments=('Lead Time', 'count'),
    Avg_Lead_Time=('Lead Time', 'mean'),
    Avg_Cost=('Cost', 'mean'),
    Avg_Sales=('Sales', 'mean')
).reset_index()
```

```
vol_threshold = route_agg['Total_Shipments'].quantile(0.75)
bottlenecks = route_agg[(route_agg['Total_Shipments'] >= vol_threshold)].sort_values('Avg_Lead_Time', ascending=False)
```

```
# Export results for summary
top_10_csv = 'top_10_efficient_routes.csv'
bottom_10_csv = 'bottom_10_inefficient_routes.csv'
ship_mode_csv = 'ship_mode_performance.csv'
bottlenecks_csv = 'geographic_bottlenecks.csv'
```

```
top_10.to_csv(top_10_csv, index=False)
bottom_10.to_csv(bottom_10_csv, index=False)
ship_mode_perf.to_csv(ship_mode_csv, index=False)
bottlenecks.to_csv(bottlenecks_csv, index=False)
```

```
print("Analysis Complete.")
print("Lead time stats:")
print(df['Lead Time'].describe())
print("\nShip Mode Performance:")
print(ship_mode_perf)
```

```
Analysis Complete.
Lead time stats:
count    10194.000000
mean      1320.841868
std        262.444892
min         904.000000
25%       1271.000000
50%       1274.000000
75%       1638.000000
max       1642.000000
Name: Lead Time, dtype: float64
```

```
Ship Mode Performance:
   Ship Mode  Total_Shipments  Avg_Lead_Time  Avg_Cost  Avg_Sales
0   First Class             1548    1338.275840    4.721124    13.772216
1   Same Day                547    1333.442413    4.411225    13.004881
2   Second Class            1979    1323.845376    4.827539    14.077928
3   Standard Class           6120    1314.334641    4.749328    13.969011
```

```
print("Top 10 Efficient Routes (by State):")
```

```

print(route_agg_sorted[['Route_State', 'Total_Shipments', 'Avg_Lead_Time', 'Lead_Time_Variability']].head(10))

print("\nBottom 10 Least Efficient Routes (by State):")
print(route_agg_sorted[['Route_State', 'Total_Shipments', 'Avg_Lead_Time', 'Lead_Time_Variability']].tail(10))

vol_median = route_agg['Total_Shipments'].median()
high_vol = route_agg[route_agg['Total_Shipments'] > vol_median]
bottlenecks = high_vol.sort_values(by='Avg_Lead_Time', ascending=False).head(10)

print("\nGeographic Bottlenecks (High Volume, High Lead Time):")
print(bottlenecks[['Route_State', 'Total_Shipments', 'Avg_Lead_Time']])

```

Top 10 Efficient Routes (by State):

	Route_State	Total_Shipments	Avg_Lead_Time \
19	Central Hub -> Maine	8	1137.125000
35	Central Hub -> Newfoundland and Labrador	6	1216.166667
29	Central Hub -> Nevada	39	1226.794872
54	Central Hub -> Virginia	224	1229.125000
41	Central Hub -> Ontario	58	1247.465517
52	Central Hub -> Utah	53	1253.000000
48	Central Hub -> South Carolina	42	1255.428571
18	Central Hub -> Louisiana	42	1263.785714
25	Central Hub -> Mississippi	53	1266.566038
42	Central Hub -> Oregon	124	1267.709677

	Lead_Time_Variability
19	0.166854
35	0.293913
29	0.195446
54	0.208620
41	0.218258
52	0.231269
48	0.192564
18	0.242745
25	0.207766
42	0.204595

Bottom 10 Least Efficient Routes (by State):

	Route_State	Total_Shipments	Avg_Lead_Time \
50	Central Hub -> Tennessee	183	1391.486339
49	Central Hub -> South Dakota	12	1395.916667
44	Central Hub -> Prince Edward Island	10	1420.300000
53	Central Hub -> Vermont	11	1438.909091
33	Central Hub -> New Mexico	37	1441.837838
15	Central Hub -> Iowa	30	1443.900000
20	Central Hub -> Manitoba	12	1455.333333
47	Central Hub -> Saskatchewan	2	1457.000000
37	Central Hub -> North Dakota	7	1637.857143
56	Central Hub -> West Virginia	4	1638.000000

	Lead_Time_Variability
50	0.178372
49	0.258146
44	0.180113
53	0.132739
33	0.185318
15	0.159153
20	0.131337
47	0.177626
37	0.000894
56	0.001221

Geographic Bottlenecks (High Volume, High Lead Time):

	Route_State	Total_Shipments	Avg_Lead_Time
50	Central Hub -> Tennessee	183	1391.486339
14	Central Hub -> Indiana	149	1381.496644
55	Central Hub -> Washington	506	1360.662055
7	Central Hub -> Connecticut	82	1357.548780
21	Central Hub -> Maryland	105	1356.780952
57	Central Hub -> Wisconsin	110	1343.000000

